

What can run together?

What must wait?

What needs protection?

Discussion draft for collaboration review, not an operating procedure.

Read this page and the summary for your subsystem. The full guide contains every one of the 40 source rules, with actions, detector stages, source disagreements and return conditions.

Situation	What we propose
LAr level is not established above the required minimum	HV is unavailable. Block voltage-setting, enabling and ramping until LAr covers all required instrumentation, including the cathode.
Camera lights or an external laser with PDS	Take turns. Block the second request and verify light-off before PDS returns.
Purity monitor with PDS	Keep PDS running and mark the affected data. PDS bias stays on for this interaction.

Define the actual PDS state. The photon detection system (PDS) team must say which powered, biased or acquiring state excludes external light. PDS-owned calibration light and power-over-fiber have their own rules.

Level first, HV availability second. A reliable measured LAr level above the specified minimum is required before voltage can be set or HV enabled. Other HV checks must also pass; reaching the level does not switch HV on.

Use the procedure for the detector stage. Installation needs work and isolation checks. Commissioning and integration need an agreed test plan and owner. A run needs the right equipment states and data-quality records.

Some answers remain open. The old PDS interface conflicts with the proposed purity-monitor rule. Cathode-loss response, power-loss ordering and several electronics conditions still need detector decisions.

[Search the complete guide by subsystem](#) · [Send a rule review](#) · [Read the findings and sources](#)

Cryogenics

HV stays unavailable until LAr covers all required instrumentation, including the cathode.

Tell the other teams: The measured LAr level and whether it is reliably above the required minimum; relevant cryogenic faults and changes that affect data quality.

Agree together: The minimum level and reference point that guarantee the required coverage; measurement reliability; the response to a falling level; and restart checks.

Related rules: [P-001](#), [P-002](#), [P-003](#), [P-004](#), [P-013](#), [P-015](#), [P-017](#), [Q-001](#), [Q-004](#).

[Open the full rules for this team](#)

High voltage and field cage

Keep voltage-setting, enabling and ramping unavailable until the required LAr level is established.

Tell the other teams: Requested and measured states, ramping, faults, affected detector region, and the reason for a shutdown.

Agree together: The approved startup and shutdown sequence, linked field-cage outputs, and the still-open response of other systems to a cathode trip.

Related rules: [S-001](#), [P-001](#), [P-002](#), [P-003](#), [P-004](#), [P-005](#), [P-006](#), [P-014](#), [P-016](#), [P-017](#), [O-001](#), [Q-003](#), [O-007](#), [H-001](#), [H-002](#), [O-009](#), [O-010](#), [P-018](#), [P-019](#), [O-011](#), [Q-005](#).

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TPC electronics and APA/CRP teams

Agree which combinations of detector states are allowed.

Tell the other teams: Actual APA/CRP bias and electronics states, readiness, cooling problems, and the detector region affected.

Agree together: Allowed states during filling and tests; response to cathode or mains-power loss; and which fan, heater or condensation problems need a shutdown.

Related rules: [P-005](#), [P-006](#), [P-007](#), [P-008](#), [P-009](#), [P-013](#), [P-014](#), [P-015](#), [P-016](#), [P-017](#), [Q-003](#), [Q-004](#), [O-004](#), [O-005](#), [O-006](#), [O-007](#), [H-001](#), [O-009](#), [O-010](#), [P-018](#), [P-019](#), [O-011](#), [Q-005](#).

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Photon detection system (PDS)

Take turns with camera lights and external lasers; keep running during purity measurements.

Tell the other teams: The exact PDS state that excludes external light, the region affected, equipment faults, and readiness to resume.

Agree together: Light-off confirmation and settling time; purity-monitor data labels; equipment protection limits; and the conflicting wording in the old PDS interface.

Related rules: [P-002](#), [P-004](#), [P-007](#), [P-008](#), [P-009](#), [P-011](#), [P-012](#), [P-016](#), [P-017](#), [O-001](#), [O-002](#), [Q-001](#), [Q-002](#), [C-001](#), [O-003](#), [Q-003](#), [Q-004](#), [O-004](#), [O-005](#), [O-006](#), [O-007](#), [H-001](#), [H-002](#), [O-009](#), [O-010](#), [P-019](#), [O-011](#), [Q-005](#).

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Cameras and illumination

Check with PDS before turning on camera lights.

Tell the other teams: Whether illumination is actually on or off, which lights are involved, and where their light can reach.

Agree together: How the second request is blocked, how light-off is confirmed, and what to do if light appears unexpectedly. The rule concerns illumination; camera power alone is not a proven conflict.

Related rules: [O-001](#), [O-003](#).

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Lasers, calibration and movable devices

Name the source and its purpose before applying a light rule.

Tell the other teams: External-laser emission and shutter state, PDS calibration activity, or device position and movement.

Agree together: The affected region, PDS handoff, laser-safety checks, permitted movement and the calibration record needed by analysis.

Related rules: [S-001](#), [S-002](#), [P-012](#), [O-002](#), [Q-002](#), [O-003](#), [O-007](#), [H-001](#), [Q-005](#).

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Purity monitors

Publish when and where a purity measurement affects PDS data.

Tell the other teams: Actual start and stop times, affected region, and whether the activity record is reliable.

Agree together: Timing precision, the affected PDS channels and any settling time. Arrange a quiet interval only when a measurement needs one.

Related rules: [Q-001](#), [Q-004](#).

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Rack power, cooling and electrical services

Protect the affected equipment and explain what lost power.

Tell the other teams: Rack or supply faults, cooling and power availability, the loads affected, and the state of protection equipment.

Agree together: Which loads switch off, which essential loads may stay on, response time, power-loss ordering and inspection before restart.

Related rules: [S-001](#), [P-006](#), [P-007](#), [P-008](#), [P-009](#), [P-010](#), [P-011](#), [P-012](#), [P-013](#), [P-014](#), [P-016](#), [P-017](#), [C-001](#), [O-005](#), [O-008](#), [O-009](#), [O-010](#), [P-018](#), [P-019](#).

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Data acquisition, timing and data quality

Know whether a run may start and which data need a label.

Tell the other teams: Run type, timing readiness, configuration identity, the detector region in use and the recorded activity intervals.

Agree together: When to stop, pause or continue a run; who can change settings; how analysis receives quality information; and fresh configuration after recovery.

Related rules: [P-002](#), [P-003](#), [P-004](#), [P-007](#), [P-008](#), [P-009](#), [P-011](#), [P-012](#), [P-013](#), [P-017](#), [O-001](#), [O-002](#), [Q-001](#), [Q-002](#), [C-001](#), [O-003](#), [Q-003](#), [Q-004](#), [O-004](#), [O-005](#), [O-006](#), [O-007](#), [H-001](#), [H-002](#), [O-008](#), [O-009](#), [O-010](#), [P-018](#), [P-019](#), [O-011](#), [Q-005](#).

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Installation, integration and run coordination

Make it clear who is using each part of the detector.

Tell the other teams: Active work, the named test owner, affected area, agreed exceptions and the handback to operations.

Agree together: The work and test procedure for each stage, exception expiry, escalation and the checks before returning to a run.

Related rules: [S-001](#), [S-002](#), [P-001](#), [P-003](#), [P-006](#), [P-007](#), [P-008](#), [P-010](#), [P-015](#), [P-016](#), [O-001](#), [O-002](#), [Q-001](#), [O-003](#), [O-006](#), [O-007](#), [H-001](#), [H-002](#), [O-008](#), [O-009](#), [O-010](#), [P-019](#), [O-011](#).

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Detector Protection System (DPS) and Slow Controls

Make the agreed action happen and show a useful reason.

Tell the other teams: Protection availability, blocked requests, actual equipment state, faults and the steps still needed for restart.

Agree together: A named subsystem owner, the affected equipment, the required response time and a test of each action. Keep required protection working through the failures it must handle.

Related rules: [S-001](#), [S-002](#), [P-001](#), [P-002](#), [P-003](#), [P-004](#), [P-005](#), [P-006](#), [P-007](#), [P-008](#), [P-009](#), [P-010](#), [P-011](#), [P-012](#), [P-013](#), [P-014](#), [P-015](#), [P-016](#), [P-017](#), [O-001](#), [O-002](#), [Q-001](#), [Q-002](#), [C-001](#), [O-003](#), [Q-003](#), [O-004](#), [O-005](#), [O-006](#), [O-007](#), [H-001](#), [H-002](#), [O-008](#), [O-009](#), [O-010](#), [P-018](#), [P-019](#), [O-011](#), [Q-005](#).

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